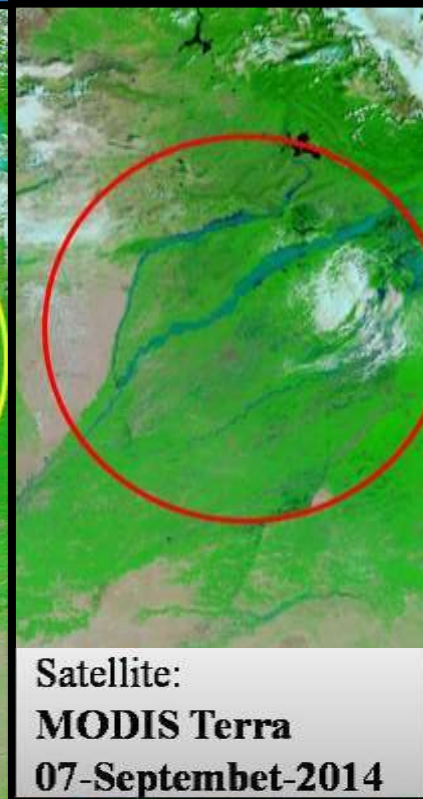


SATELLITE DATA APPLICATION IN DISASTER MONITORING



Learning Objectives

To introduce with examples the application of satellite data and geospatial techniques for hazard planning and risk management

Satellite data for natural hazards. Examples are

LANDSAT satellites- the longest running exercise in collecting multispectral data (Landsat 1 launched in 1972) provide necessary information for studying land surface change.

Sentinel-1 SAR: Radar observations operate day and night and penetrate cloud cover, supporting flood-extent mapping and earthquake-related ground-deformation analysis.

Sentinel-2 MSI: Optical imagery with 13 spectral bands at 10–60 m resolution supports mapping of water, vegetation, snow, and land-cover change. Useful applications include glacial-lake monitoring and burn-scar assessment

RADARSAT: An efficient and effective satellite sensor for assessing flood damages and oil seeps.

MODIS Terra and Aqua, with daily coverage, can be used for rapid disaster mapping.

Sentinel-2 MSI: Optical imagery with 13 spectral bands at 10–60 m resolution supports mapping of water, vegetation, snow, and landcover change. Useful applications include glacial-lake monitoring and burn-scar assessment.

continue...

Satellite data for natural hazards. Examples are

VIIRS active-fire data: Near-real-time thermal hotspot detections at 375 m resolution complement MODIS and support wildfire monitoring.

GPM IMERG: Multi-satellite precipitation estimates at half-hourly intervals support assessment of extreme rainfall, flood-triggering conditions, and drought.

OPERA Dynamic Surface Water Extent: Sentinel-1-derived water maps at 30 m resolution support monitoring of changing surface-water extent. Flood assessment requires comparison with normal or pre-event water conditions.

NASA OPERA DSWx-S1 Copernicus Global Flood Monitoring: Automatically processes Sentinel-1 radar observations to provide worldwide satellite-based flood monitoring.

And many more...

Pakistan Flood 2010



Qambar Shahdadkot

01 Jun 2010



27 Aug 2010



Guddu Barrage

Pre Flood – June 02, 2010



Post Flood – August 12, 2010



Flood 2012

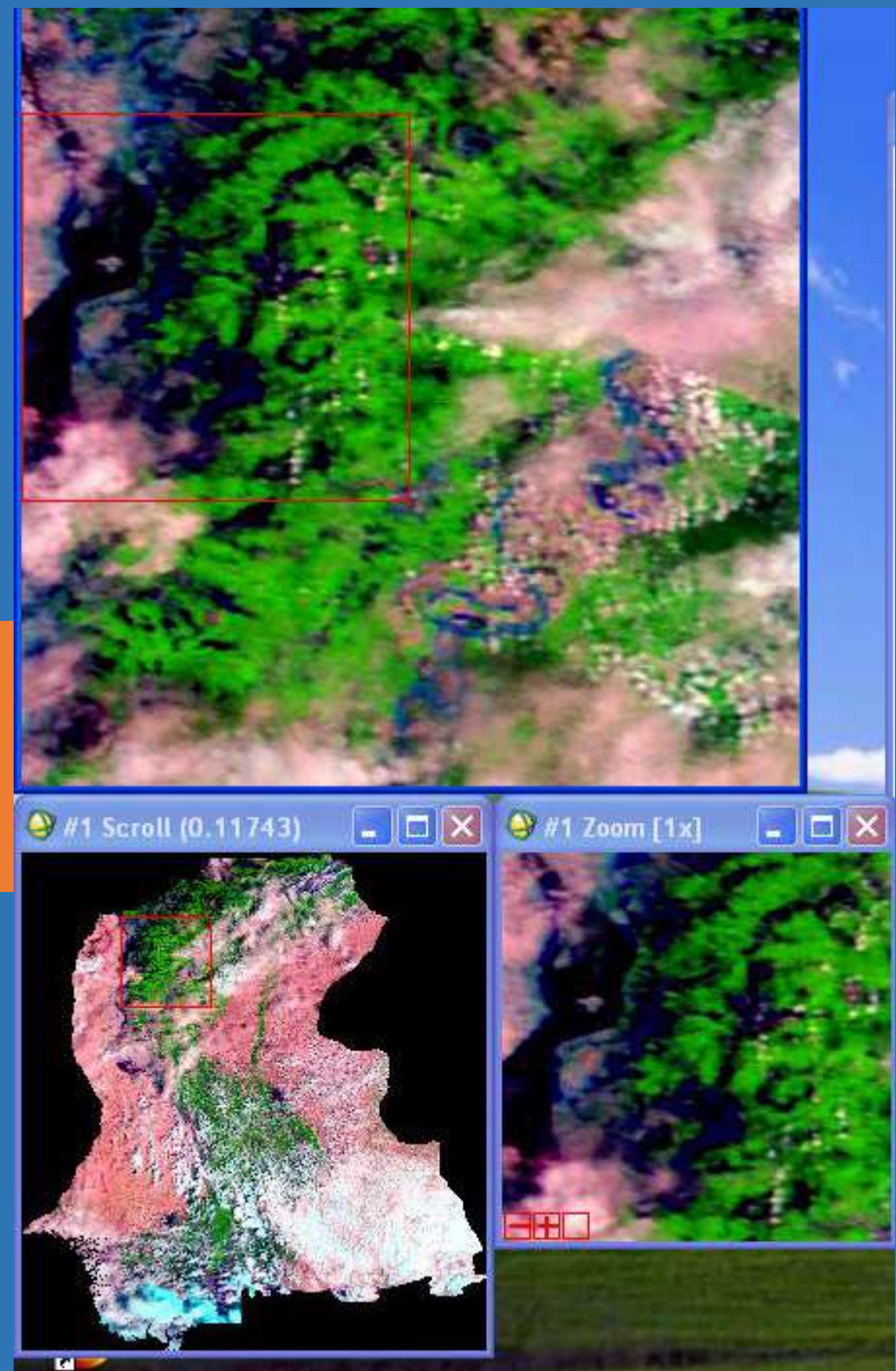
Pakistan Pre Flood Map: May 30, 2012



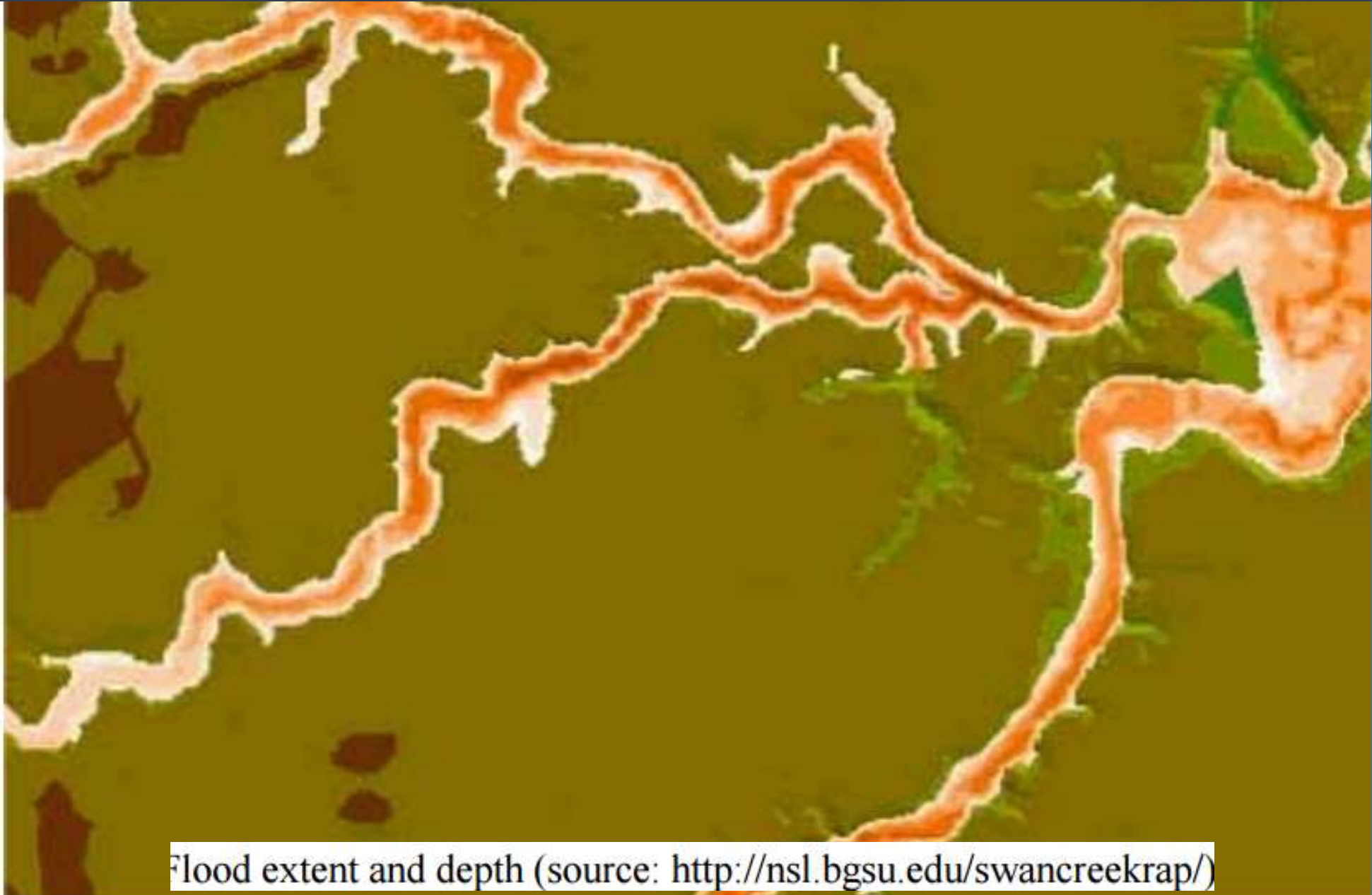
Pakistan Flood Map: September 13, 2012



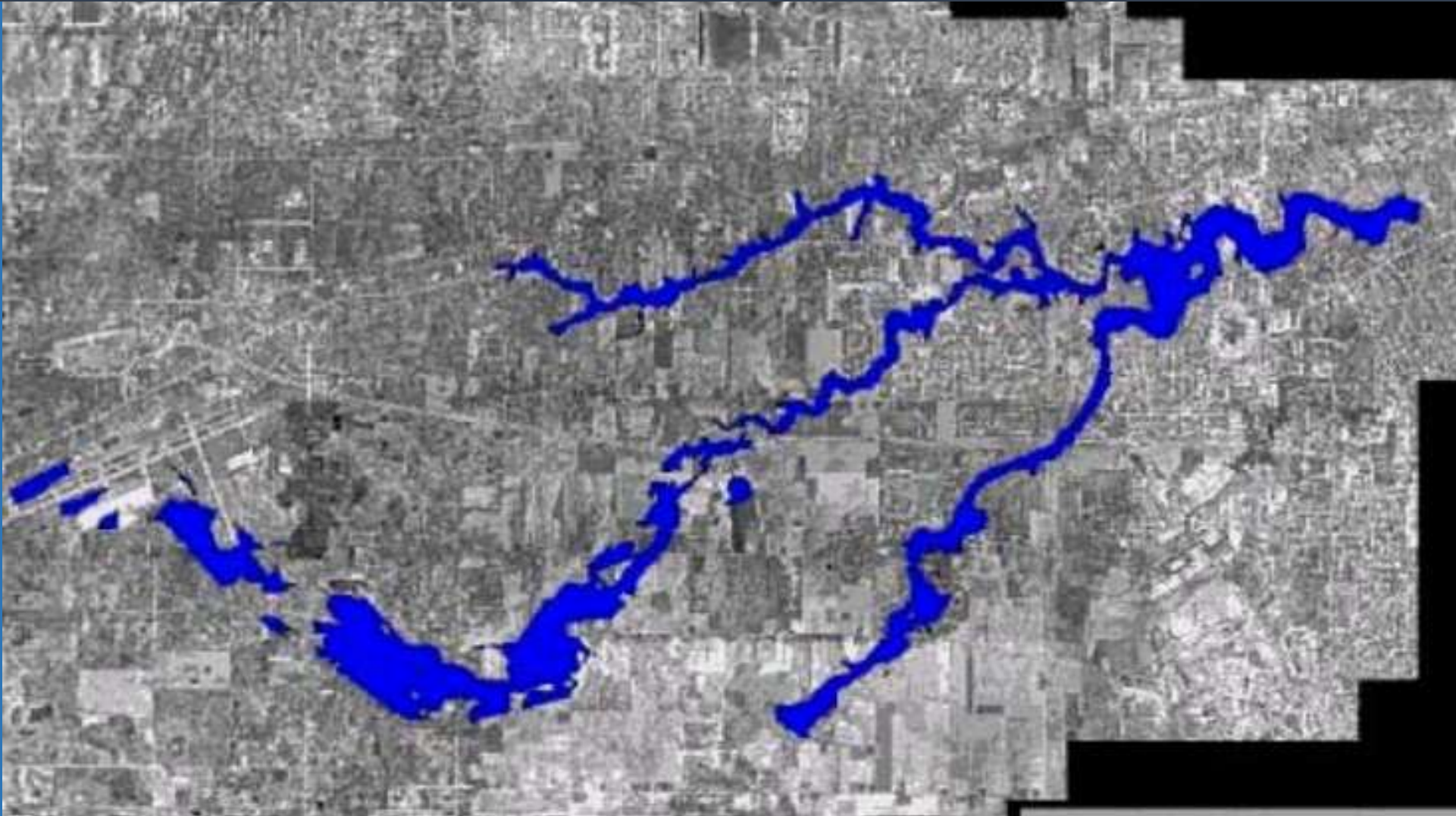
MODIS Image Showing Flood Extent
on
Sept 13, 2012



Flood Modeling using Satellite data and GIS tools

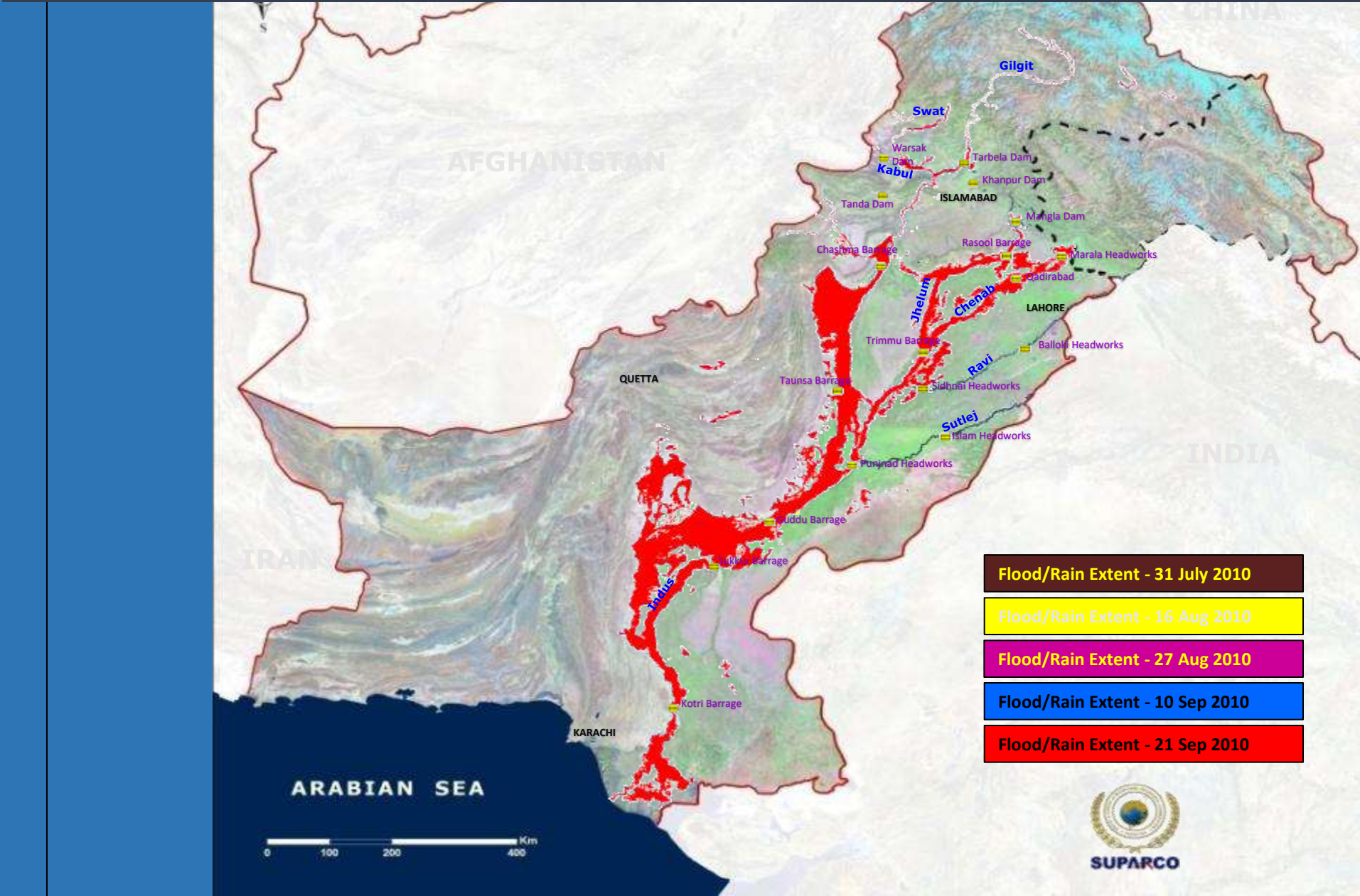


Damage Assessment using Satellite data and GIS tools

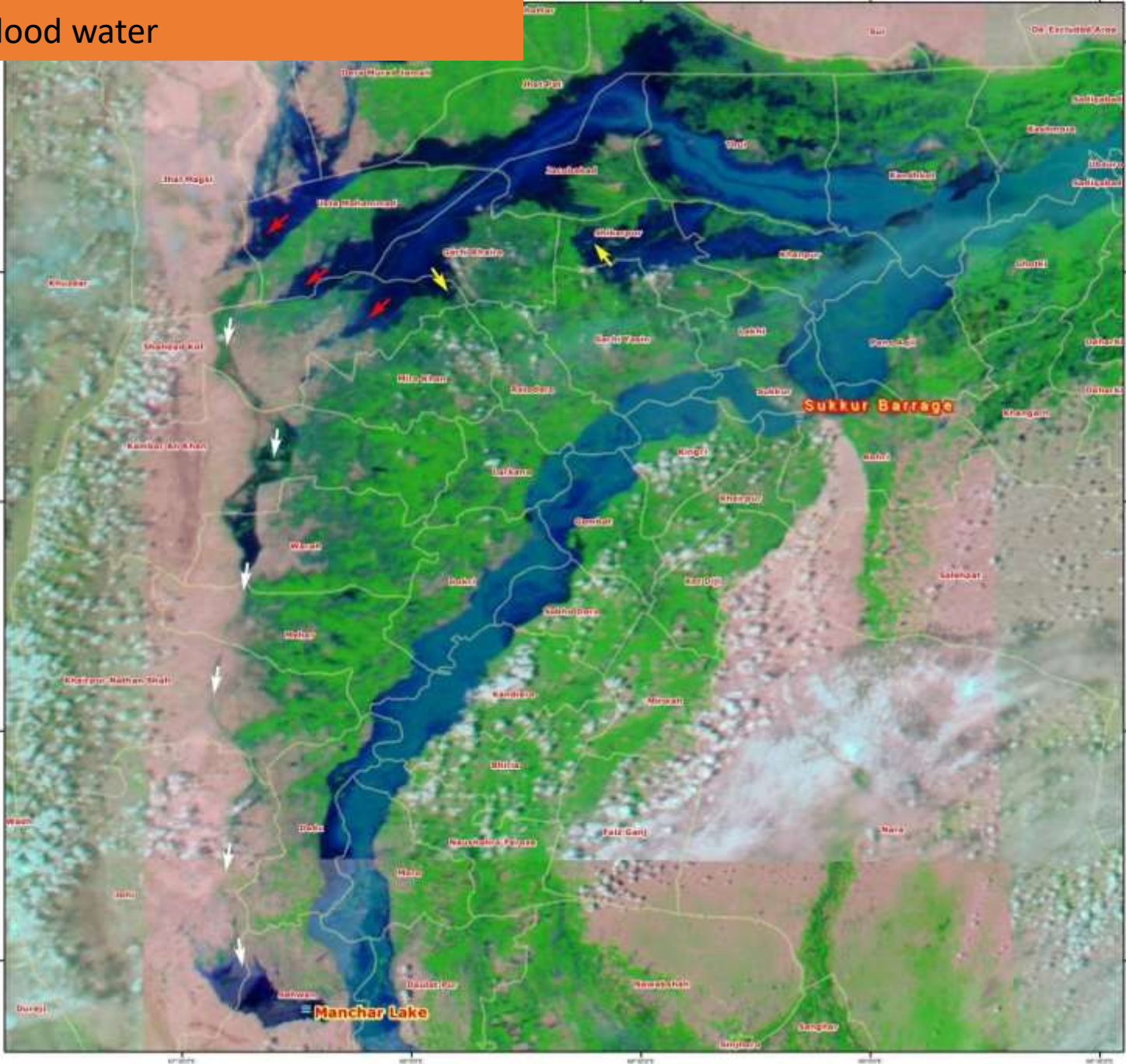


Flooding Extent overlay on Elements at Risk
(source: <http://nsl.bgsu.edu/swancreekrap/>)

Rapid Flood Mapping by SUPARCO



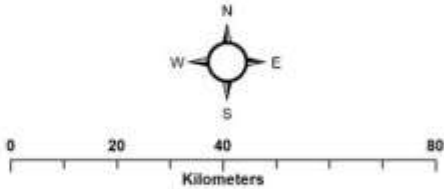
Direction of Flood water



Legend

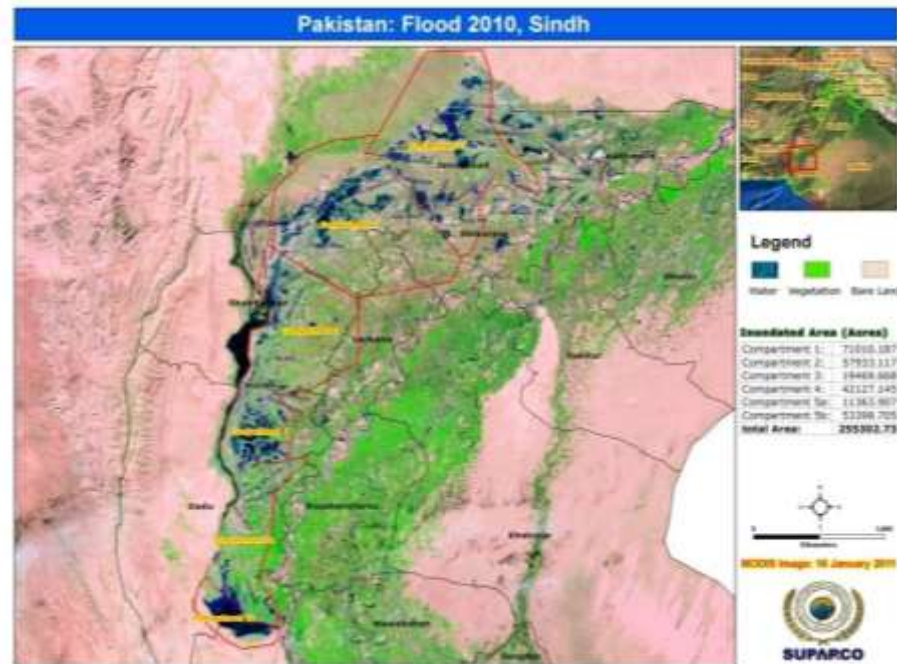
-  Dam / Barrage
-  Tehsil Boundary
-  Fast Movement
-  Slow Movement
-  Potential Movement

PRODUCED ON 21 AUGUST 2010

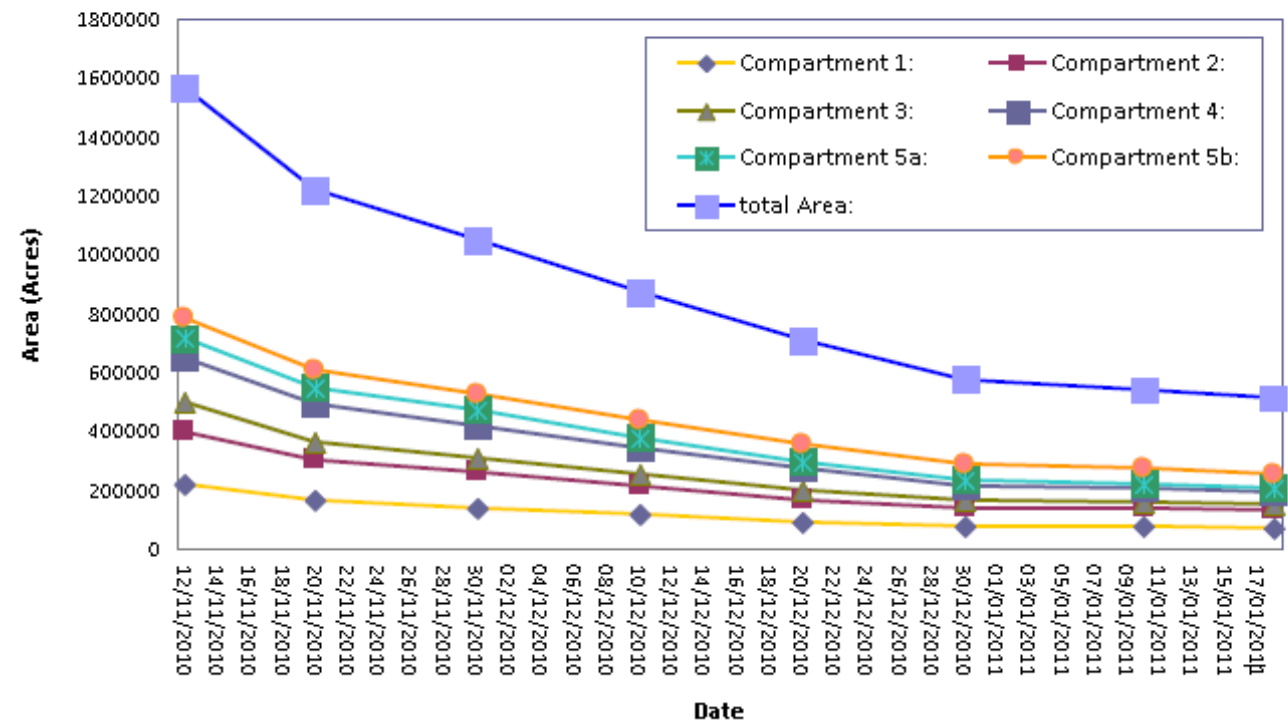


Sindh Dewatering Program

Daily flood extent maps produced by utilizing data from remote sensing satellites



Graphical Representation of Flood Water Reduction in Sindh



Pakistan Floods — 2025

- VIIRS false-color images show flooding along major rivers.
- Compare September 9, 2025 with the same date in 2023.
- GIS task: overlay inundation with settlements and cropland.

Image Link: [NASA Earth Observatory \(2025\)](#)

Flood Woes Continue in Pakistan

IMAGE OF THE DAY FOR OCTOBER 14, 2022

Water has begun to recede, but large swaths of farmland remain waterlogged as food shortages loom and waterborne diseases spread.

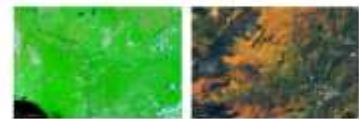
NASA Earth Observatory

OCT 14, 2022

ARTICLE



VIEW MORE IMAGES OF THE DAY:



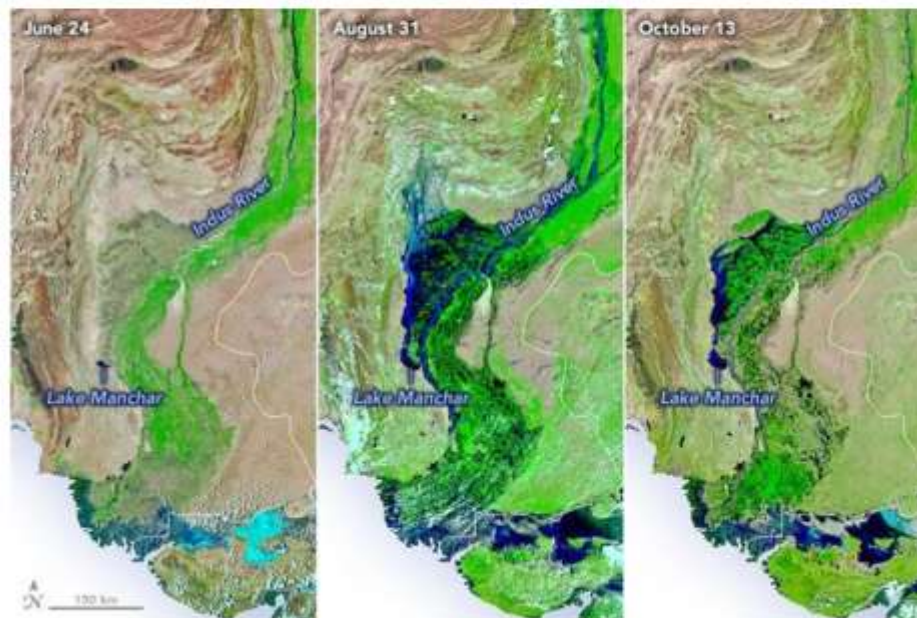
OCT 13, 2022

OCT 15, 2022

INSTRUMENTS:

NOAA-20 — VIIRS

Model



June 24–October 13, 2022

In early September 2022, floods in Pakistan were the worst in a decade. Monsoon rains had pummeled the region for several weeks and floodwaters inundated 75,000 square kilometers of the country. Six weeks later, rains have ceased, and fields have begun to drain. But vast swaths of farmland remain waterlogged, infectious diseases are spreading, and food shortages loom.

The images above show the progression of the flooding. The second image shows Sindh province on August 31, 2022, near the peak of the flooding. By October 13, 2022 (third image), a considerable amount of water had drained off the landscape and back into rivers. But many areas remained wet and waterlogged in comparison to June 2022 (first image). All three images were acquired by the Visible Infrared Imaging Radiometer Suite (VIIRS) on the NOAA-20 satellite. They are false-color images based on VIIRS observations of shortwave infrared and visible light, a combination that makes it easier to distinguish between water (blue) and land (green).

Valencia Flash Floods — 2024

- Landsat 8 observed inundation on 30 October 2024.
- Urban land, farms, and coastal wetlands were affected.
- GIS task: compare flood extent with drainage and transport routes.

Image Link: [NASA Earth Observatory \(2024\)](#)



Satellite Image before Cyclone



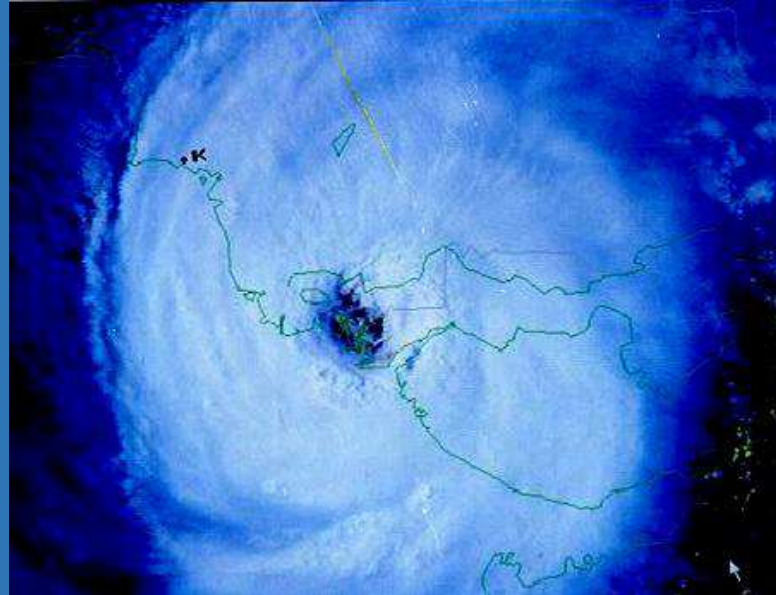
Mangrove Forests deteriorated due to Cyclone 02A



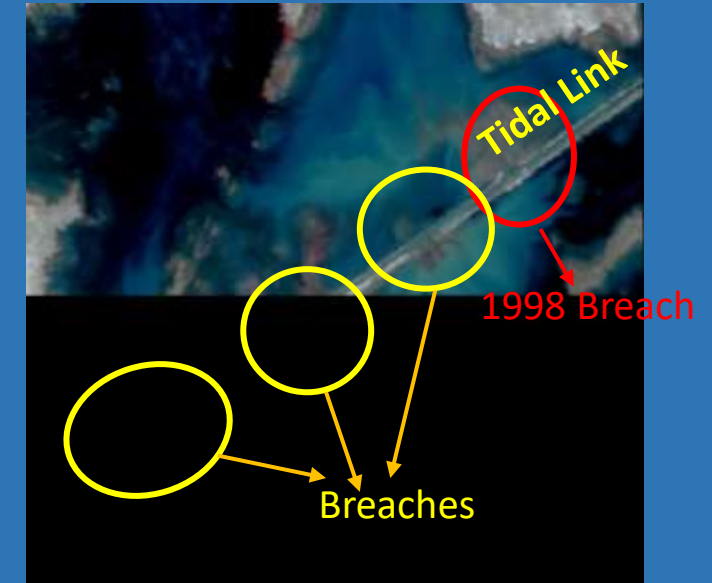
Satellite Image after Cyclone



Prior cyclone Satellite Image of Feb.1999 of Tidal Link Area



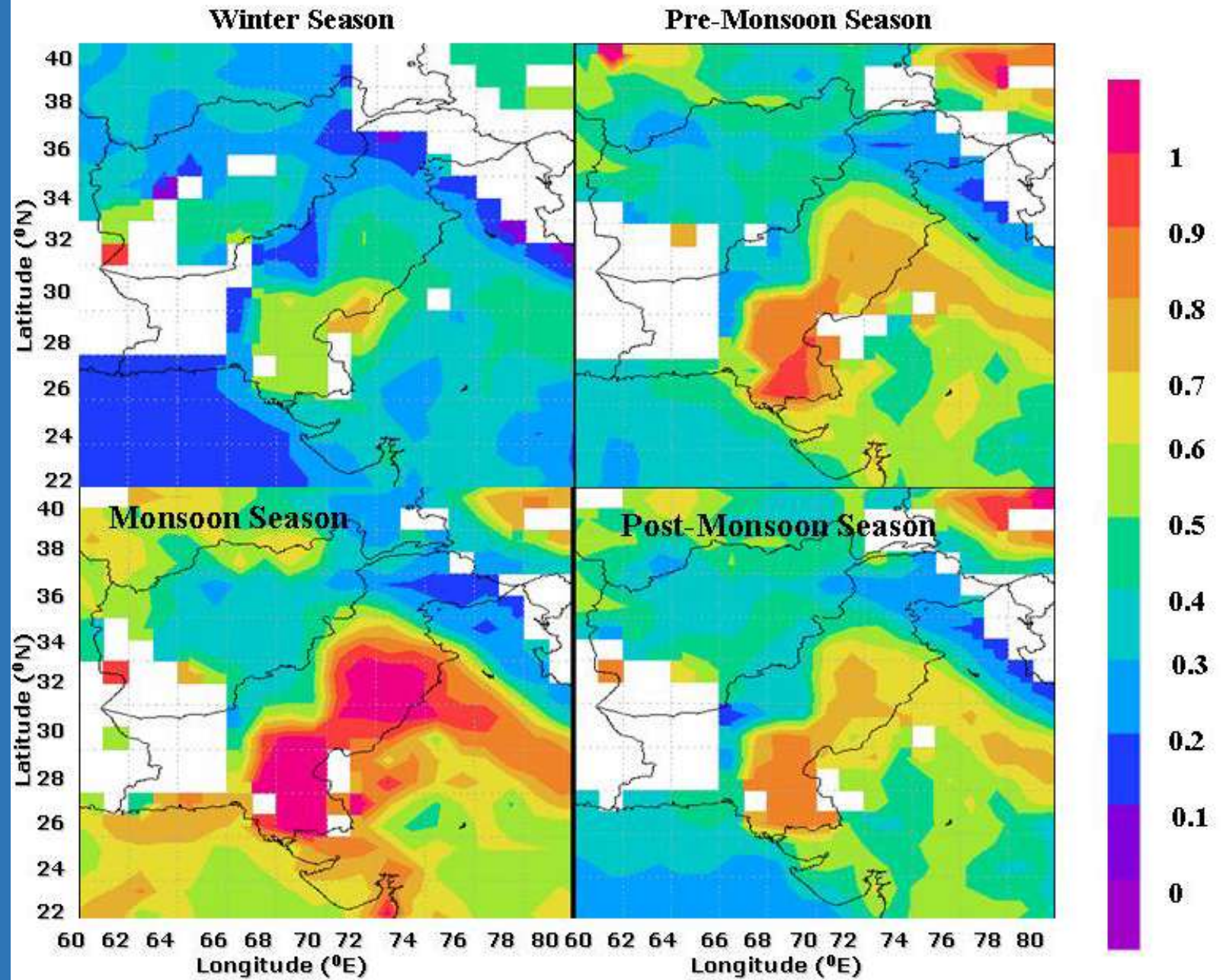
Satellite picture of powerful Cyclone TC 02A, hitting the Thatta District on May 20th,1999



Post cyclone Satellite Image Nov. 1999 of Tidal Link Area

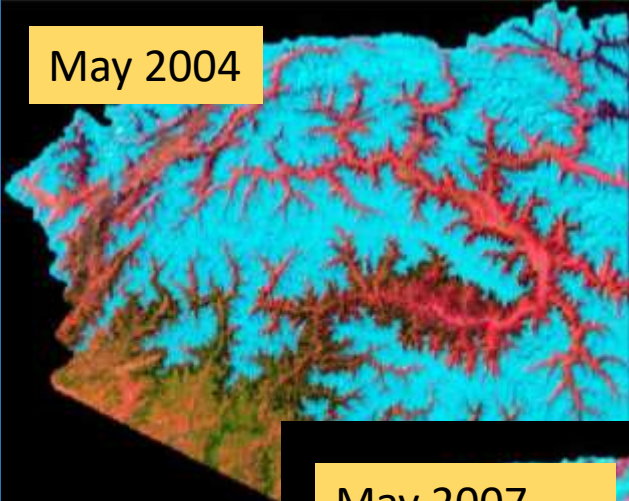
Aerosol Optical Depth (AOD)

2006 - 2007

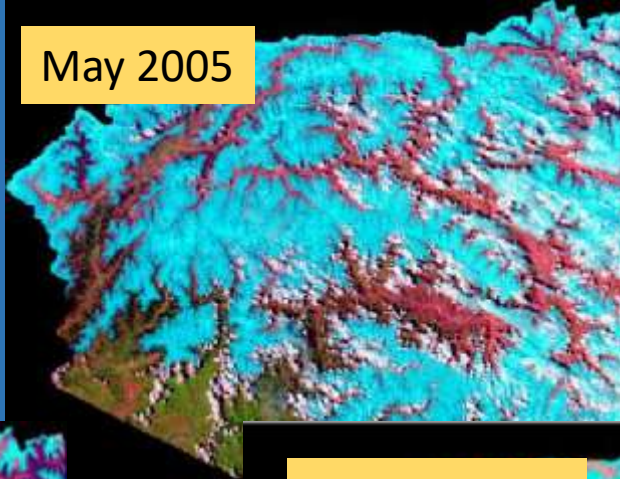


Snow Cover Mapping

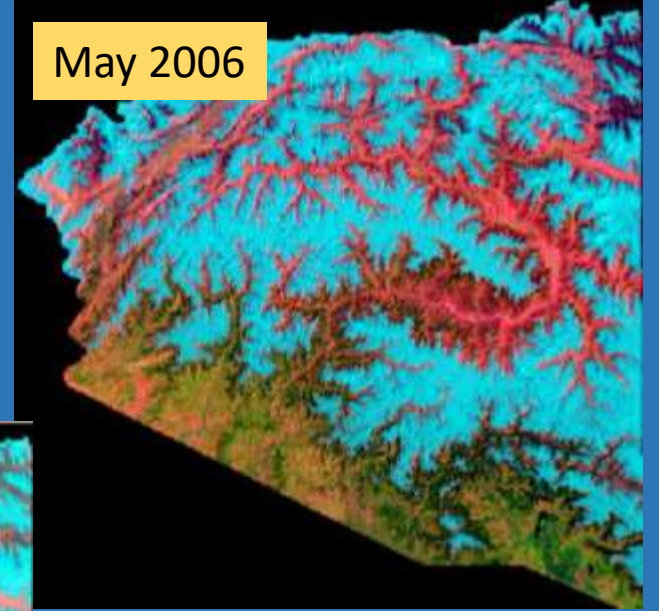
May 2004



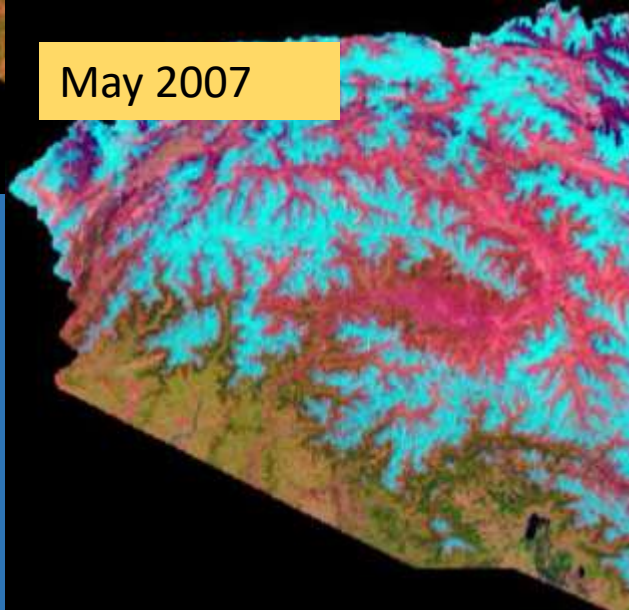
May 2005



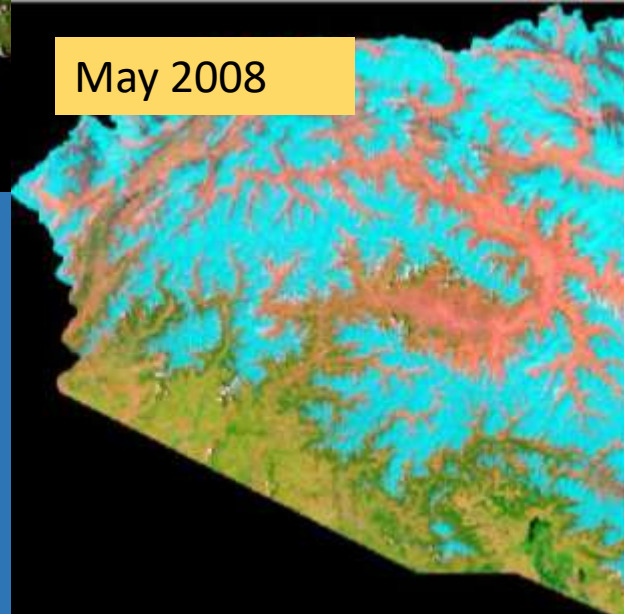
May 2006



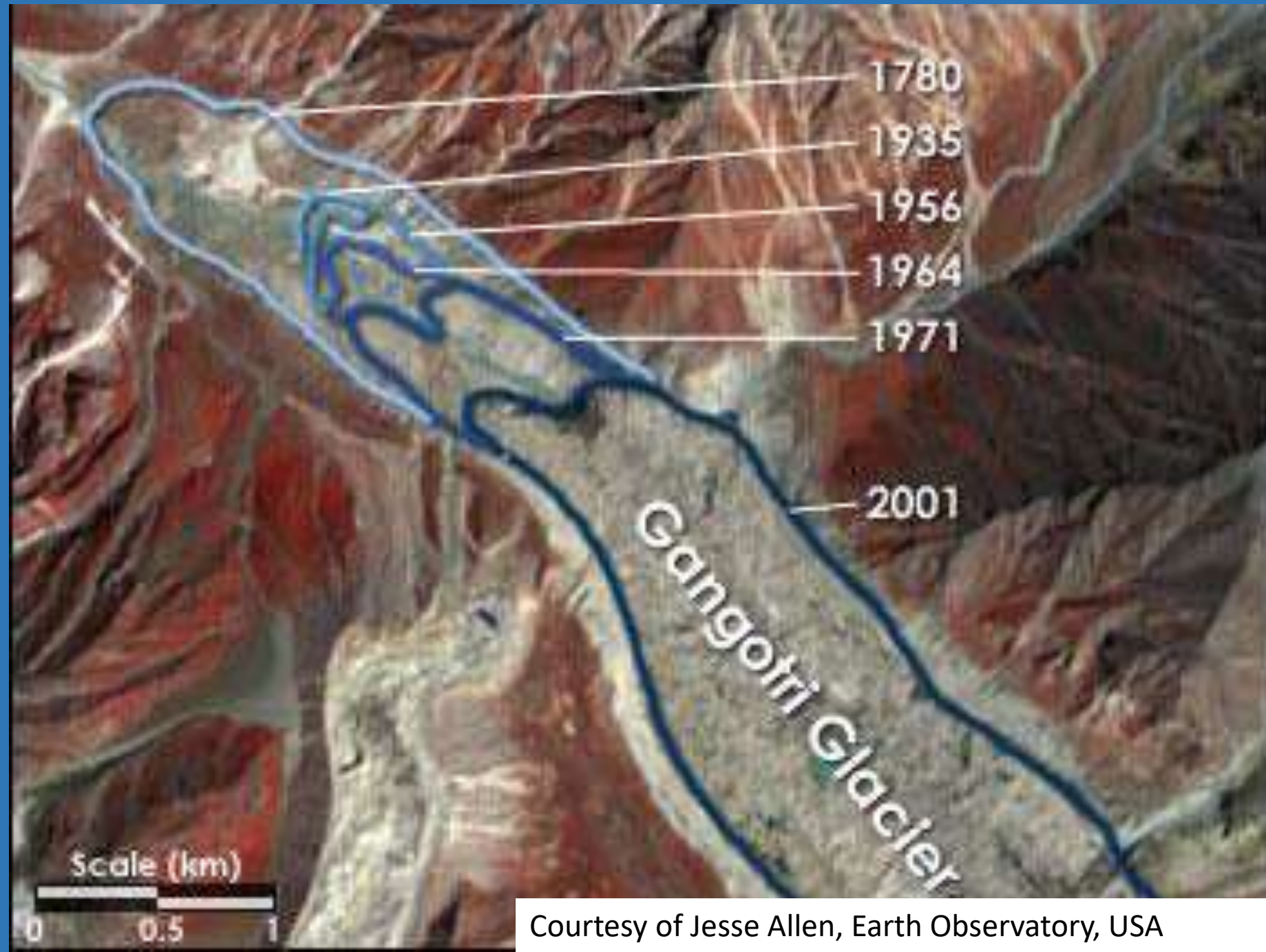
May 2007



May 2008

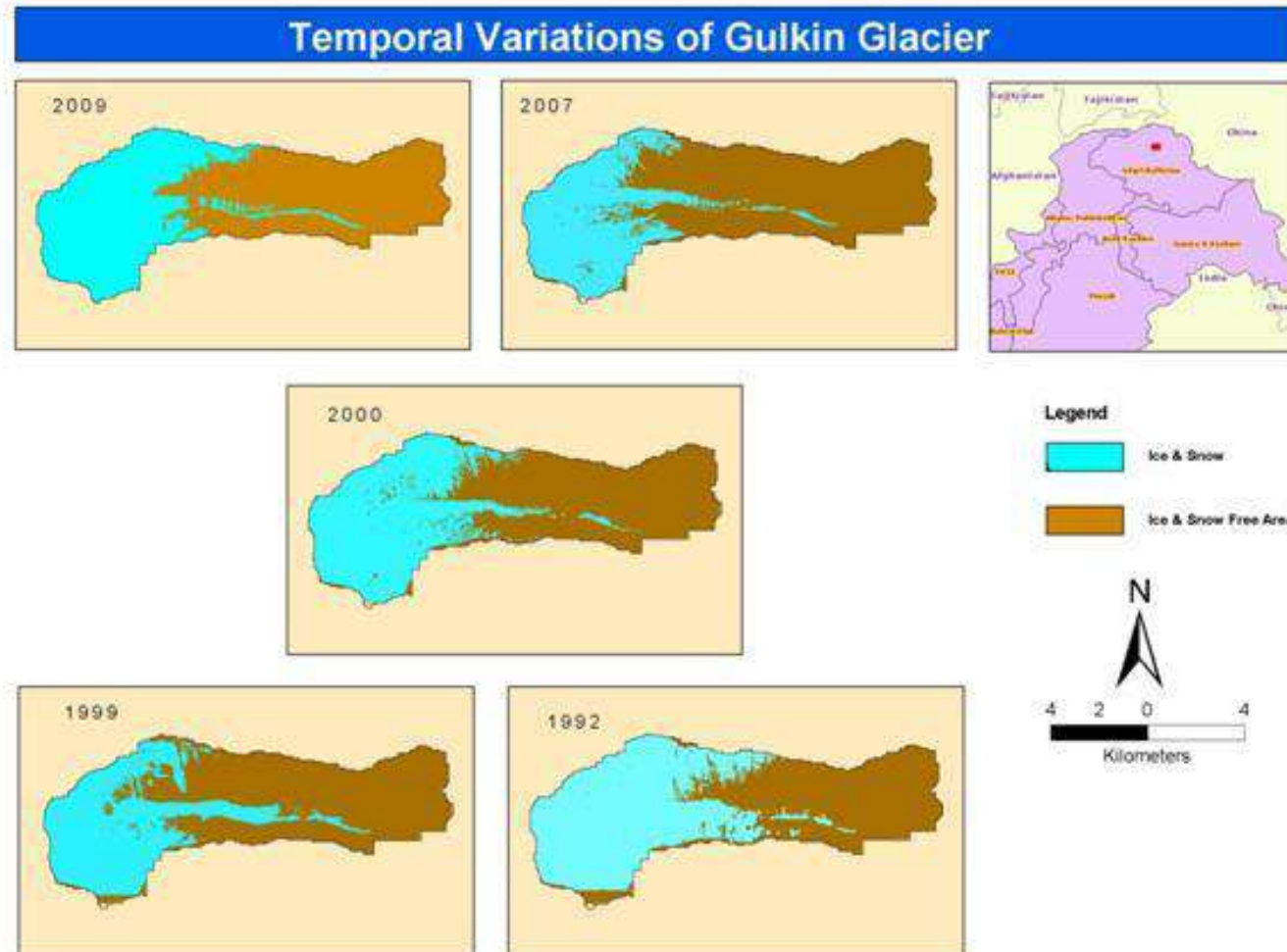


Estimates of Gangotri
(India) Glacier Retreat
(ASTER Image)

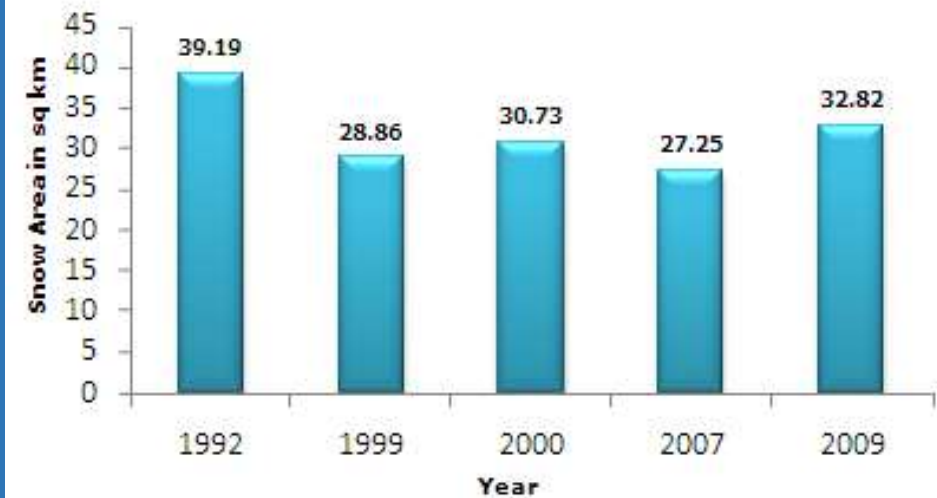


Courtesy of Jesse Allen, Earth Observatory, USA

Annual Variation in Glacial Snow Cover



Temporal changes in the Snow Cover area of Gulkin Glacier



Thame GLOF, Nepal — 2024

- On 16 August, a rock avalanche triggered two lake outbursts.
- Satellite images, drone surveys, and fieldwork reconstructed the event.
- Lesson: Small lakes can produce large, cascading hazards.

Image Link: [ICIMOD / NDRRMA report \(2025\)](#)

Thame: Mapping Downstream Risk

- Map the lakes, unstable slopes and connected river channels.
- Overlay settlements, bridges and hydropower facilities.
- Use field checks and modelling to assess evacuation options.

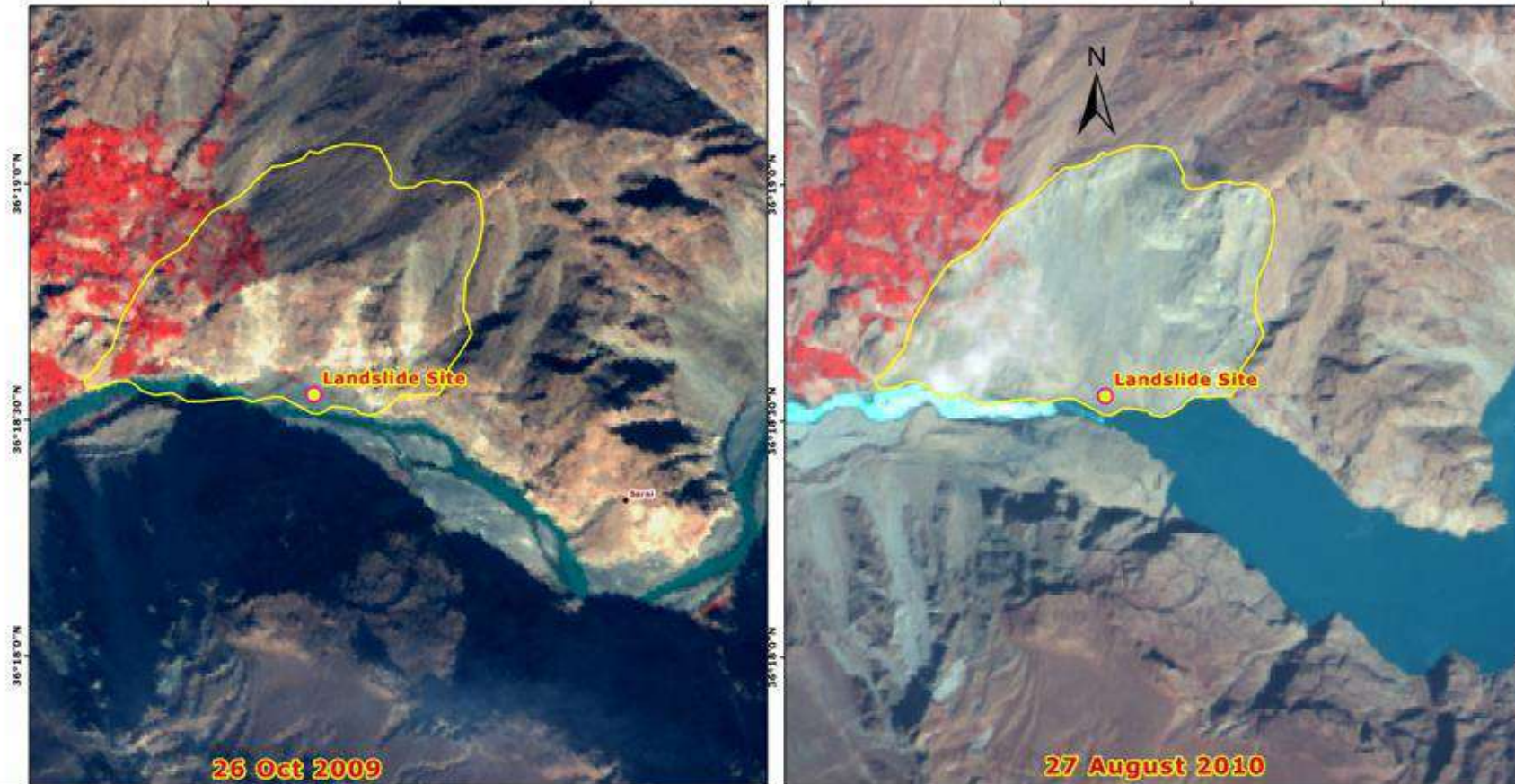
Image Link: [ICIMOD: Thame study \(2025\)](#)

Rasuwa Flood, Nepal — 2026

- A major flood struck the Nepal-China border on 26 August.
- Assess upstream changes and exposed infrastructure downstream.
- Cause assessment is preliminary: do not assume a confirmed GLOF.

[ICIMOD event updates \(2026\)](#)

Pakistan: Hunza Landslide Extent



Landslides in Muzaffarabad AJK

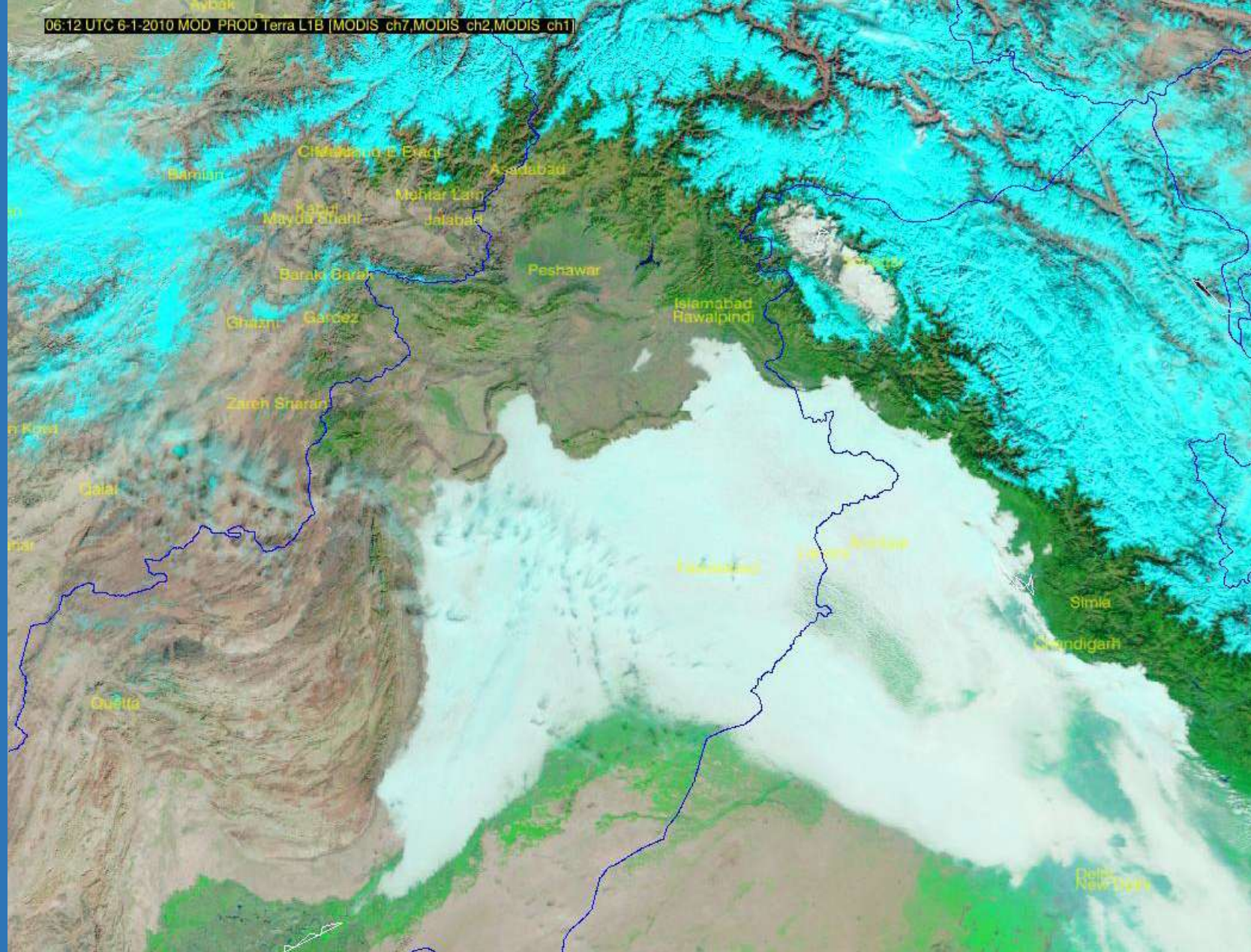


Türkiye-Syria Earthquakes — 2023

- Sentinel-1 radar measured earthquake-related ground displacement.
- High-resolution optical images supported damage assessment.
- GIS task: map affected buildings and usable access routes.

Image Link: [European Space Agency \(2023\)](#)

06:12 UTC 6-1-2010 MOD PROD Terra LTB [MODIS_ch7,MODIS_ch2,MODIS_ch1]



Winter Fog Over Pakistan - MODIS Image

Satellite Aided Search and Rescue Program



Search & Rescue



- Constellation of Russian, US, French and Canadian Geo-stationary & LEO Satellites
- Receives Distress Signals from Aircrafts, Vessels, Personnel
- 121.5 MHz, 243 MHz and 406 MHz Beacons

Data Reception



Ground Receiving Station

Data Distribution



Rescue Coordinate Centers

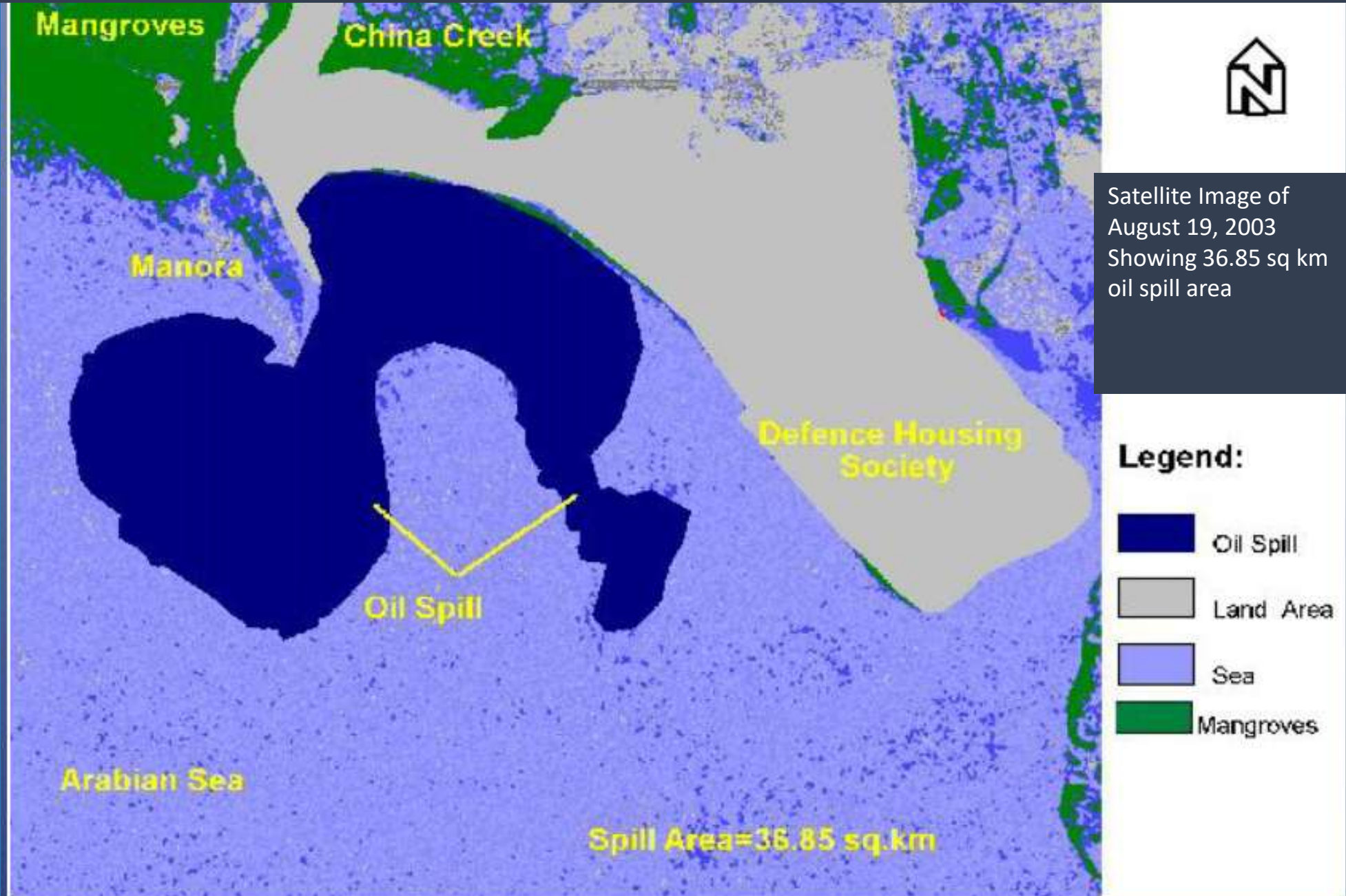


Data Product

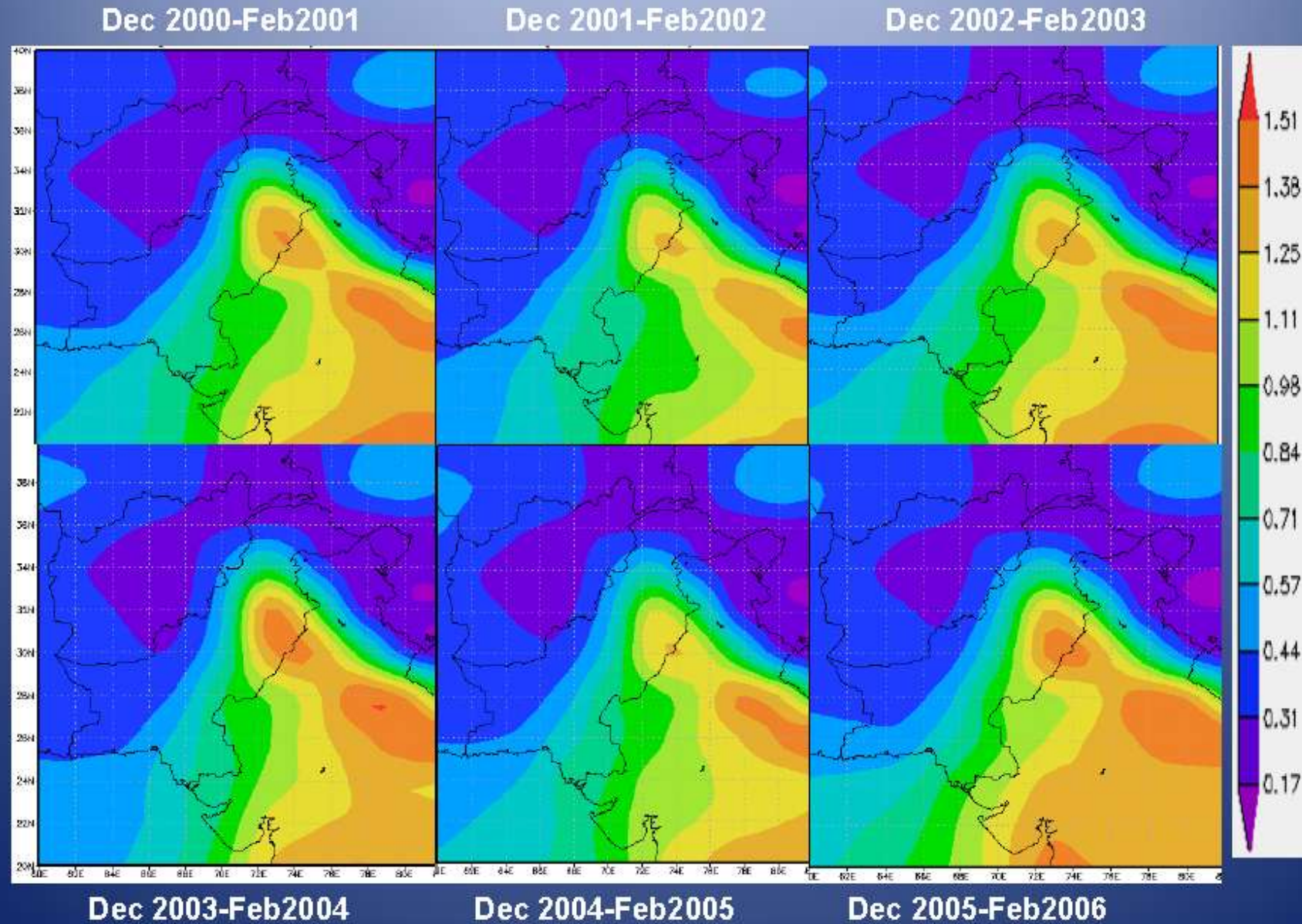


Data Processing

Oil Spill - RADARSAT DATA (Microwave RS)



Black Carbon: Column Optical Depth $\{ (10^{-2}) \text{ Unit les}$ (Winter Season)



Los Angeles Wildfires — 2025

- Satellite observations tracked the January fires and their impacts.
- Sentinel-1 change maps flagged potentially damaged structures.
- GIS task: prioritize field inspection of buildings and access routes.

Image Link: [NASA Applied Sciences \(2025\)](#)

From Satellite Evidence to Decisions

- Remote sensing shows where conditions have changed.
- GIS identifies the people and infrastructure exposed.
- Field evidence, uncertainty, and vulnerability guide action.

Discuss: which map supports which decision?

QUESTIONS?